Exercise 8.2.9. (i) Show that the category of finite abelian groups or finite dimensional $k[x]$-modules does not contain nonzero projective objects (so it does not have enough projectives).

(ii) Let A be a finitely generated commutative ring. Show that the category of finitely generated A-modules has enough projectives.

Problem 8.2.10. Let V be a finite dimensional vector space over a field k. Let $C_i = SV \otimes \wedge^i V$. We can view C_i as the space of polynomial functions on V^* with values in $\wedge^i V$. Define the differential $d_i : C_i \to C_{i-1}$ by the formula

$$d_i(f)(u) = \iota_u f(u), \quad u \in V^*,$$

where $\iota_u : \wedge^i V \to \wedge^{i-1} V$ is the contraction defined by the formula

$$\iota_u(v_1 \wedge \cdots \wedge v_i) = \sum_{j=1}^{i} (-1)^j u(v_j) v_1 \wedge \ldots \hat{v}_j \cdots \wedge v_i.$$

(i) Show that $C_\bullet$ is a projective (in fact, free) resolution of k (with trivial action of V) as an SV-module. (It is called the **Koszul resolution**.)

(ii) Let $V = U \oplus W$. Then we can view SW as an SV-module (U acts by zero). Construct a resolution of SW by free SV-modules whose terms are $C_i = SV \otimes \wedge^i U$.

Hint: Tensor the Koszul resolution of k as an SU-module by SW.

(iii) Regard SV as an $S(V \oplus V) = SV \otimes SV$-module using left and right multiplication. Construct a free resolution of SV as an $SV \otimes SV$-module with terms $C_i = SV \otimes \wedge^i V \otimes SV$ (called the Koszul bimodule resolution).

(iv) By tensoring the resolution of (iii) over SV with any SV-module M, construct a projective (in fact, free) resolution $P_\bullet$ of M with $P_i = 0$ for $i > \dim V$. Deduce that for any SV-module N and any $i > \dim V$, one has

$$\mathrm{Tor}_i^{SV}(M, N) = \mathrm{Ext}_{SV}^i(M, N) = 0$$

(the **Hilbert syzygies theorem**).

(v) Compute $\mathrm{Ext}_{SV}^i(k, k)$ and $\mathrm{Tor}_i^{SV}(k, k)$.